

Business Intelligence Systems

MGTA457, Fall 2025

PROFESSOR: Greg Schibler

EMAIL: gschibler@ucsd.edu

PHONE: 858-212-2952

OFFICE HOURS: By appointment, (Rady 3s133 or zoom)

TEACHING ASSISTANT: TBD

EMAIL: tbd

DESCRIPTION

This is a hands-on, project-based course in which you will learn to transform today's high-volume data from disparate systems into real-time, readily actionable, business insights via highly interactive charts, maps, and graphs. We will be using one of the top BI tools in the industry, Tableau, but the principles of good data visualization and dashboard design will be readily transferable to any modern BI tools. In addition to developing a very marketable skill, your finished project will make a nice addition to a work product portfolio.

OBJECTIVES

At the close of this course, you will be able to:

- Articulate the value of BI systems
- Employ the principles of good infographic design to convey information rapidly and accurately
- Recognize when design may lead to poor interpretation or even be designed to deceive (a nice life skill)
- Evaluate related BI technologies
- Identify data sources and plan BI Implementation
- Combine, cleanse, and group data feeds for dashboard creation
- Design and build an effective, interactive dashboard
- Present effectively, using data story views

MATERIALS

Required

- Laptops running Windows 8 or newer, or Mac OS Mojave or later, for use in class
- Software: Tableau License Key will be supplied during first class session and in Canvas. (You will not need it installed for the first class session, but please have it installed and working prior to our second class meeting.)
- Tableau whitepapers and eBooks on Dashboard Design Principles and data hygiene to be provided on Canvas

Recommended

No textbook is required for this course; however, the following are great additional sources covering the fundamentals of this course, and I've drawn heavily from Steve Wexler's book for the content of our first class meeting:

- Steve Wexler (2021): The Big Picture: How to Use Data Visualization to Make Better Decisions—Faster; ISBN-10: 126047352X
- Cole Nussbaumer Knaflic (2015): Telling with Data: A Data Visualization Guide for Business Professionals; ISBN-10: 1119002257

SCHEDULE

Class Date	Class Topic & Activities	Assignments (due before class)
Week 6 (November 5 th , 2025)	Course Overview, Intro to BI, Big Data Landscape, Principles of Effective Visualizations and Dashboards	
Week 7 (November 12 th , 2025)	Hands-On Tableau: Creating your first simple, but highly effective dashboard	• Read: Visual Analysis Best Practices • Read: How to design a dashboard (pgs 1-39) • Teams chosen in Canvas
Week 8 (November 19 th , 2025)	Hands-On Tableau: Dynamic data grouping and filtering, mapping, calculations and parameters	• Read: How to design a dashboard (pgs 40-78) • Read: 10 Best Practices for Building Effective Dashboards • Presentation design document due, 1 per team
Week 9 (November 26 th , 2025)	Hands-On Prep Builder: Data ETL: Merging, reformatting, pre-processing, and cleansing	• Read: Tidy Data in Tableau Prep
Week 10 (December 3 rd , 2025)	Student team Dashboard Presentations	• Presentation in class • Final Project Documentation Due • All Dashboard Files Due
Finals Week (December 10 th , 2025)	Student team Dashboard Presentations	• Peer Reviews Due • Team Reviews Due

ASSIGNMENTS

Reading Assignments

All readings will be supplied in Canvas

Team Projects

Small student teams will work together to design and implement a story of informative dashboards that combine multiple data sources, and teams will use their dashboards to tell a story in their final presentation. The group project will comprise four components, and there will be one deliverable for each team:

1. *Dashboard Design Document*: This is a 2-5 page document with simple sketches of expected elements and the intended interaction between those elements. The purpose of this document is to thoughtfully consider the business use of the dashboard, the end-user's expected capability and needs, and then apply the principles of good design, prior to developing any views or dashboards.
2. *Tableau's Prep Builder Workflow*: will integrate two or more datasets, with at least four appropriate cleaning components.
3. *Dashboard and Presentation*: At least 3 dashboards with 6 or more views (each student on the team should individually create at least two views) will be used to tell a story about the data analyzed in a classroom presentation of 10-15 minutes, and answer questions from reviewers. (extra credit is available for good use of storyboard)
4. *Final BI Project Document*: Less than 5 pages of text, along with any appropriate images, to describe the project goals, team roles, data sources and prep, dashboard interactions and findings. Additionally, enhancements from the Design Document should be considered relative to business goals, intended audience or useability.

Peer and Team Presentation Reviews

Each student will complete a short survey of each other team member, relative to quantity of work, value of work, and degree of team co-operation.

Each student will attend four other team presentations, and provide simple feedback on those teams per a rubric to be provided.

GRADING

Assignments	Percentage of Final Grade
In Class Quizzes	5
Dashboard Design Document	10
Tableau Data Prep Workflow	20
Dashboard and Presentation	45
Final BI Project Document	20
Peer and Team Reviews and Class Participation	+/- 10

COURSE POLICIES

Punctuality: Class will start on time, both at the beginning of class and after breaks. You are expected to be present and ready to participate at that time, and it counts toward the participation grade. This is particularly important as we will periodically be doing exercises in break-out groups. If you are unable to attend a class, please reach out to me in advance so that we can set up an additional assignment as a substitute for your attendance. Regardless, you will still need to submit all assignments on-time, and you should also view any missed class's recording link available in Canvas.

Generative AI: You will be expected to use Generative AI tools like ChatGPT+, Co-pilot, Bing, etc. in this class. If you do not, you will likely be at a disadvantage compared to your peers in this class. Learning to use Generative AI tools is critical for everyone working in Business Analytics and Data Science.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD Website (<https://osd.ucsd.edu>) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.